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CSA5117

Cryptography and
network security

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Assignment - 1

1) => Encrypt the message "DEFEND THE WALL" using
Caesar cipher with key=3.

=> Decrypt the message "KHQR ZRUQ" using
Caesar cipher with key=3.

=> A Caesar cipher is used to encrypt the word
"SECRET" to "VHFUHW". Determine the key used,
and decrypt "WKLV LV D WHVW".

Encrypt "DEFEND THE WALL"

For encryption

$$c = (p + k) \bmod 26$$

Each letter is shifted 3 position forward;

Plaintext	D	E	F	E	N	D	T	H	E	W	A	L	L
+3 Shift	G	H	I	H	Q	Q	W	K	Z	D	D	O	O

DEFEND THE WALL

GHIHQQ WKH ZDOO

b) Decrypt "KHOO R ZRUOG" using key=3.

For decryption:-

$$P = (C - k) \bmod 26$$

Shift every position 3-letters backward:

Cipher text	K	H	O	O	R	Z	R	U	O	G
-3 shift	H	E	L	L	O	W	O	R	L	D

KHOO R ZRUOG

HELLO WORLD

c) Find the key and decrypt "WKLV LV D WHVW".

Given:-

SECRET $\rightarrow$ VHFUHW

compare:

$$S \rightarrow V = +3$$

$$E \rightarrow H = +3$$

$$C \rightarrow F = +3$$

$\therefore$ key = 3.

WKL V $\rightarrow$ THIS

LV $\rightarrow$ IS

D $\rightarrow$ A

WHVW $\rightarrow$ TEST

key = 3.

Plaintext: THIS IS A TEST

2) $\Rightarrow$ using the substitution key;

A $\rightarrow$ Q, B $\rightarrow$ W, C $\rightarrow$ E, D $\rightarrow$ R, E $\rightarrow$ T, F $\rightarrow$ Y, ..., Z $\rightarrow$ M

Encrypt the word "FIRE".

$\Rightarrow$ Given this cipher text "UJHHD QEB NRFZH YOLTK
and knowing it is encrypted with a monoalphabetic
substitution. Find the plaintext using frequency
analysis.

a) Encrypt "FIRE":-

Given substitution key;

A $\rightarrow$ Q

B $\rightarrow$ W

C $\rightarrow$ E

D $\rightarrow$ R

E $\rightarrow$ T

F $\rightarrow$ Y

...

Z $\rightarrow$ M

We need:-

F $\rightarrow$ Y

I $\rightarrow$ I

R $\rightarrow$ R

E $\rightarrow$ T

FIRE $\rightarrow$ YIRT

b) Decrypt "UJHHD QEB NRFZH YOLTK"

using frequency analysis and common English word pattern:-

QEB $\rightarrow$ THE

The phrase:

UJHHD QEB NRFZH YOLTK

Answer:-

THE QUICK BROWN FOX?

A standard monoalphabetic substitution example:-

QEB NRFZH YOLTK CLU GRJMP LSBO QEB IXWV

ALD

which decrypts to:

THE QUICK BROWN FOX JUMPS OVER THE LAZY
DOG.

Conclusion:-

The given cipher text "UJHHO QEB NRFZH YOLTK" cannot be uniquely and consistently decrypted as written without additional information or a corrected ciphertext.

- 2) $\Rightarrow$ Encrypt the message "ATTACKATPALACE" using the keyword "LEMON".
 $\Rightarrow$ Decrypt the cipher text "LXFOPVEFRNHR" keyword "LEMON".
 $\Rightarrow$ Given the encrypted message "QNXEPKMAEGTU" using the vigenere cipher and keyword "KEY", Find original word:-

a) Plain text: ATTACKATPALACE

Key: LEMONLEMONLEMO

convert letters to numbers

use: A=0, B=1, C=2, ..., Z=25

Encryption Formula:-

$$C = (P + k) \bmod 26$$

Plaintext	A T T A C K A T P A L A C E
key	L E M O N L E M O N L E M O
Cipher	L X F O P V E F D N W E O S

Answer:

ATTACKATPALACE $\rightarrow$ LXFOPVEFDNWEOS.

b) Decrypt "LXFOPVEFRNHR" using keyword "LEMON".

key: LEMONLEMONLE

Decryption formula:

$$P = (C - k) \bmod 26$$

Ciphertext	L X F O P V E F R N H R
key	L E M O N L E M O N L E
Plaintext	A T T A C K A T D A W N

Answer:-

LXFOPVEFRNHR $\rightarrow$ ATTACKATDAWN.

c) Decrypt "QNXEPKMAEGTU" using keyword "KEY".

Repeat key:

K E Y K E Y K E Y K E Y

Use:

$$P = (C - k) \bmod 26$$

Ciphertext	Q	N	X	E	P	K	M	A	E	G	T	U
key	K	E	Y	K	E	Y	K	E	Y	K	E	Y
Plaintext	G	J	Z	U	L	M	C	W	G	W	P	W

Result:-

Q N X E P K M A E G T U $\rightarrow$ G J Z U L M C W G W P W

This does not produce meaningful English Plaintext, indicating that the cipher text or key in the question may contain an error.

4) Given a 2×2 key matrix: $k = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

$\Rightarrow$ Encrypt the message "HI" using Hill cipher.

$\Rightarrow$ If the "PO" was encrypted to "QC" using the above Hill cipher matrix.

a) Encrypt "HI"

H = 7, I = 8

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$C = kP$$

$$= \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 21 + 24 \\ 14 + 40 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

$$45 \bmod 26 = 19 \rightarrow T$$

$$54 \bmod 26 = 2 \rightarrow C$$

Answer:

$$HI \rightarrow TC$$

$\Rightarrow$ verify "PO" $\rightarrow$ "QC"

$$P=15, Q=14$$

$$P = \begin{bmatrix} 15 \\ 14 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix}$$

$$= \begin{bmatrix} 45 + 42 \\ 30 + 70 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

$$87 \bmod 26 = 9 \rightarrow J$$

$$100 \bmod 26 = 22 \rightarrow W$$

$$PO \rightarrow JW$$

conclusion:

$\Rightarrow$ it does not produce QC using the commonly used matrix.

$\Rightarrow$ Therefore, The exact matrix is required to verify PO $\rightarrow$ QC

5) => Encrypt the message "MEETMEATERLUNCH" using keyword "ZEBRAS"

=> Decrypt the message "EAMTEFCUHETREL", using keyword "BOARD".

keyword: ZEBRAS

Alphabetical order:

ABERSZ

original positions:

Letter	Z	E	B	R	A	S
column no.	6	3	2	4	1	5

Write plain text row-wise:-

Z	E	B	R	A	S
M	E	E	T	M	E
A	F	T	E	R	L
U	N	C	H	X	X

Cipher text:-

MRXETCEF NTEHELXMAU

without spaces:-

MRXETCEF NTEHELXMAU

Decrypt "EANMTEFCU HETREL" Using "BOARD".

keyword: BOARD

Alphabetical order:-

A B D O R

Column order:-

A = 3

B = 1

D = 5

O = 2

R = 4

Cipher length = 15

number of rows: $15/5 = 3$

Arrange cipher text into columns in alphabetical order:

A	B	D	O	R
E	A	N	M	T
E	F	C	U	H
E	T	R	E	L

Rearrange columns to original keyword BOARD:-

B	O	A	R	D
A	M	E	T	N
F	U	E	H	C
T	E	E	L	R

Answer:

AMETNPFUEHCTEELR

6) $\Rightarrow$ using the one-time pad key XMCKL, encrypt the message "Hello".

$\Rightarrow$ why the message "ATTACKNOW" encrypted using OTP with key "QWERTYUIO" is unbreakable under ideal condition.

USE: A=0, B=1, ..., Z=25.

Encryption:-

$$C = (P + k) \bmod 26$$

Plaintext	H	E	L	L	O
value	7	4	11	11	14
key	X	M	C	K	L
key value	23	12	2	10	11
Sum mod 26	4	16	13	21	25
Cipher text	E	Q	N	V	Z

Answer:-

HELLO + XMCKL $\rightarrow$ EQNVZ.

Why is OTP unbreakable under ideal conditions?

Plaintext = ATTACKNOW

key = QWERTYUIO

A one-time pad is theoretically unbreakable when;

=> The key is truly random.

=> The key is as long as the message

=> The key is used only once.

=> The key is kept completely secret.

Therefore, an attacker can not determine the original plaintext from ciphertext alone.

OTP is unbreakable under ideal conditions because the ciphertext provides no statistical information about plaintext truly random, secret, equal length key is used only once.

7) Encrypt the message:-

Plain text: "WE ARE DISCOVERED"

keyword: "ZEBRA".

keyword : ZEBRA

Alphabetical order : A B E R Z

Column order : 5 3 2 4 1

Write message row-wise:-

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

A → EOD

B → ASR

E → EIE

R → RCE

Z → WDV

Ciphertext : EOD ASR EIERCE WDV

Decrypt "EVLNEACD TKESQAQ ROFOJDEECUXWN" using keyword "TRIAL".

keyword : TRIAL

order : AILRP

Column order : 4 3 5 2 1

Number of columns = 5

$$28 \div 5 = 5 \text{ remainder } 3$$

Encrypt "INFORMATIONSECURITY" using keyword "HACK".

Length: 19

keyword: H A C K

Number of columns: 4

rows required: $\lceil 19/4 \rceil = 5$

Arrange in rows:-

H	A	C	K
I	N	F	O
R	M	A	T
I	O	N	S
E	C	U	R
I	T	X	X

Here x is added as filler letter.

A $\rightarrow$ N M O C T = N M O C T

C $\rightarrow$ F A N U Y = F A N U Y

H $\rightarrow$ I R E I = I R E I

K $\rightarrow$ O T S R X = O T S R X

Final ciphertext:-

N M O C T F A N U Y I R E I O T S R X

Number of rows:-

5

Padding:-

1 padding character X.

8) you are given following cipher text using a monoalphabetic substitution cipher: Assume it an english sentence.

Cipher text:-

UZQSOVVOH X MDPVG PZPEVSG ZWSZOPFPESXUW

ETUCG

The most frequently occurring letters

$\Rightarrow O$

$\Rightarrow U$

$\Rightarrow Z$

$\Rightarrow P$

$\Rightarrow S$

In english, common letters are,

E, T, A, O, I, N, S, H, R.

For example:-

$O \rightarrow E$

$U \rightarrow T$

$Z \rightarrow A$

O $\rightarrow$ E

U $\rightarrow$ T

Z $\rightarrow$ A

P $\rightarrow$ O

Answer:-

"Additional frequency analysis and pattern matching" to determine the exact substitution table.

q) you intercepted a message encrypted with a monoalphabetic substitution cipher.

ciphertext: BJSW QJW YMJW NX F XNBUQJWX.

Compare the known words:-

Plaintext

ciphertext

THIS

BJSW

IS

QJ

THE

YMJ

But the mapping conflict:

For example:-

T $\rightarrow$ B

H $\rightarrow$ J

I $\rightarrow$ J.

conclusion:-

The cipher text:-

BJTW QJW YMJW NX F XNRUQJWX

can not be correctly decrypted using the known plaintext "this is THE". under monoalphabetic substitution cipher, because it creates conflicting substitution mapping.

1) cipher text:- ZICVTWQNGRZGV TWAVZHCQ YGLMGJ

The key is unknown and is between 2 and 5 letters.

Repeated sequences in the ciphertext can help estimate the key length.

Therefore, the key cannot be reliably recovered from the supplied ciphertext alone without performing systematic computational cryptanalysis.

METHOD:-

- => Test key length 2, 3, 4 and 5.
- => Perform frequency analysis on each group.
- => Determine a Caesar shift for each group.
- => Combine the shifts to recover Vigenere key.
- => Decrypt the cipher text using recovered key.

Therefore, the cipher text requires systematic Vigenere cryptanalysis to recover the unknown key. the supplied data does not allow a reliable unique key answer through straightforward manual calculation alone.

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